ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CIT5241 INFORMATION RETRIEVAL AND WEB SEARCH

FINAL EXAMINATION

2nd JUNE 2025

14:00 – 17:00

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks.**
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Scenario:

You are evaluating three information retrieval systems (System A, System B, System C) on a test collection with 10 queries. For each query, relevance judgments (on a scale of 0 to 3, where 0 = not relevant, 3 = highly relevant) and the top-5 retrieved documents are provided.

Example Data for Query Q1:

- Relevance Judgments:
    - Document D1: 3
    - Document D2: 2
    - Document D3: 1
    - Document D4: 0
    - Document D5: 0
- Retrieved Results (Top-5):
    - System A: [D1, D3, D5, D2, D4]
    - System B: [D2, D1, D4, D3, D5]
    - System C: [D3, D5, D1, D2, D4]

Assume similar data is available for all 10 queries.

a) For Query Q1 only , compute the following metrics for each system at retrieval depth 5:

i. Precision@5 [3 Marks]

ii. Recall@5 [3 Marks]

iii. Average Precision (AP) [6 Marks]

iv. Normalized Discounted Cumulative Gain (nDCG@5) [6 Marks]

b) For all 10 queries, compute:

i. Mean Average Precision (MAP) [6 Marks]

ii. Which system performs best based on MAP? Justify. [3 Marks]

iii. Would nDCG@5 change the conclusion? Explain. [3 Marks]

c) Discuss the importance of incorporating performance metrics that penalize retrieval systems for placing non-relevant documents higher in rank than relevant ones. In your answer, explain how such metrics influence the effectiveness of information retrieval systems in meeting users' information needs. Support your discussion with examples of specific metrics that address this issue. [20 Marks]

(subtotal: 50 Marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

a) Test collection-based evaluation is a standard methodology in information retrieval (IR) to assess the effectiveness of retrieval systems. The evolution of test collections in information retrieval (IR) evaluation began with the Cranfield experiments in the 1950s and 1960s, which established foundational methods using fixed sets of documents, queries, and relevance judgments. This framework was significantly expanded by the Text REtrieval Conference (TREC) starting in 1992, which introduced large-scale, diverse collections and standardized evaluation protocols, facilitating broad IR research collaboration. Subsequent developments saw the creation of specialized collections for domains such as web, and multimedia.

i. Discuss with the aid of a suitable diagram, the process of generating relevance judgements for test collections [15 Marks]

ii. Discuss the challenges that still exist for the test collection based evaluation methodology. Your discussion should also include attempts made by the research community in information retrieval to address these challenges.

[10 Marks]

Question 3

a) Below is Figure 1 showing how two human judges rated the relevance of a set of twelve documents to a particular information need (0 = non-relevant, 1 = relevant). Let us assume that you've written an IR system that for this query returns the set of documents {4, 5, 6, 7, 8}.

docID	Judge 1	Judge 2
1	0	0
2	0	0
3	1	1
4	1	1
5	1	0
6	1	0
7	1	0
8	1	0
9	0	1
10	0	1
11	0	1
12	0	1

Fig 1. Relevance of twelve (12) documents rated by human judges

i. Calculate the kappa measure between the two judges [10 Marks]

ii. Calculate precision, recall, and F1 of your system if a document is considered relevant only if the two judges agree. [15 Marks]

Question 4

i. Describe with the aid of an example the procedure employed in the vector space model to generate rankings of documents given a particular query. [5 Marks]

ii. Using your answer in (i), generate a ranked list given the query and the list of documents below. *(show working)*

Query: "algorithm search retrieval"

Document 1: "The algorithm improves search efficiency in retrieval systems."

Document 2: "Search algorithms are widely used in information retrieval"

Document 3: "Search algorithms are a popular research topic today."

Document 4: "Search algorithms must be both effective and efficient."

[20 Marks]

(25 marks)

END OF EXAMINATION